

Final assignment
Microsystems Design and Modelling (ET4260)
Part 3/3

Instructions:

Document your calculations properly with derivations and screenshots when needed.

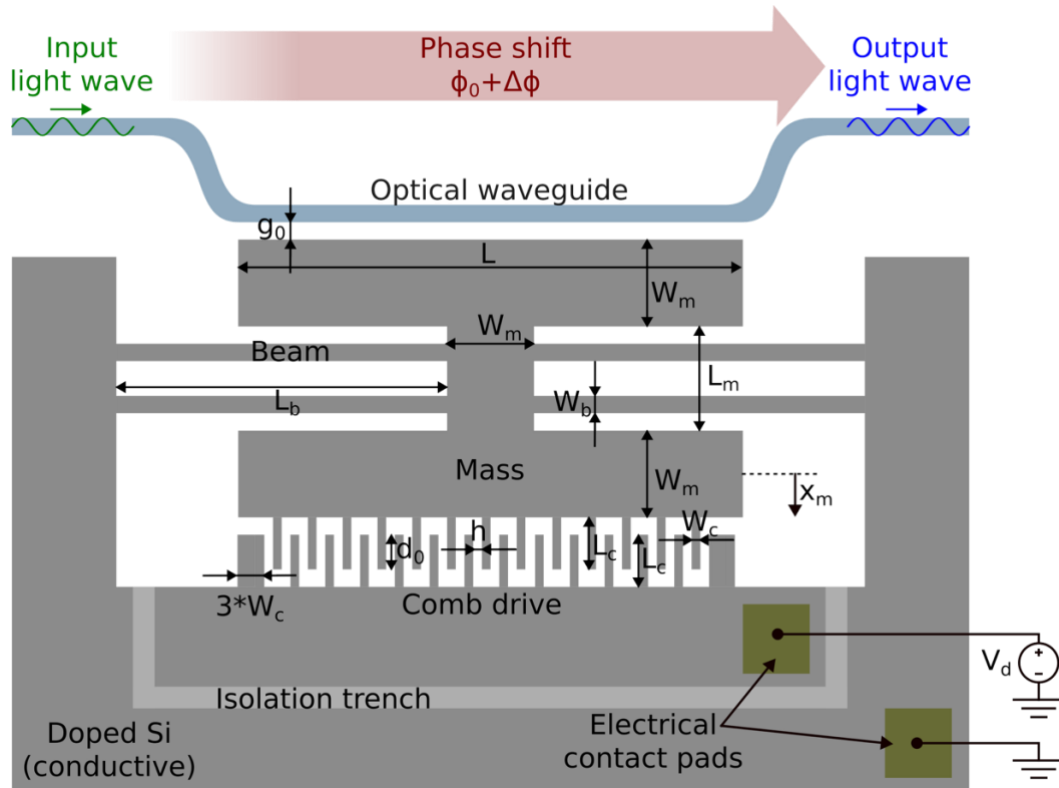
Please submit your files in word format to facilitate feedback. Latex-generated PDF files are also accepted but not preferred. **We expect type-set equations and derivations. Images of handwritten work will NOT be graded.**

Deadline for final report submission (including all parts: part 1 + part 2 + part 3): 24th June 2026

Please, mark corrections to part 1 and part 2 made after the first submission in **blue**.

State here your NAME and STUDENT NUMBER.

In this final part, consider the same system described in Part 1 and Part 2 (image and design parameters are on page 2). The technology available for device fabrication has **allowed all dimensions to be scaled to the smallest feature size of 200 nm**. (This includes gaps and widths, as shown in the table above). Your task is to now update the geometry of the modulator (feature sizes minimum 200 nm, maximum 1 mm, $2 < N < 200$) to improve the modulation sensitivity (part a) and mitigate out-of-plane behavior (part b):



*Some dimensions are not drawn on scale for better visualization.

Feature	Symbol	Part1/2 Dimension	Part 3a Restrictions	Final Value
Length of modulation section	L	100 μm	Min 200 nm Max 1 mm	
Modulation gap at rest	g_0	200 nm		
Length of beams	L_b	75 μm		
Width of beams	W_b	5 μm		
Length of mass central section	L_m	40 μm		
Width of mass sections	W_m	30 μm		
Comb-drive gap	h	1 μm		
Comb-drive overlap length at rest	d_0	19 μm		
Comb-drive finger length	L_c	20 μm		
Comb-drive finger width	W_c	1 μm		
Bottom Comb-drive fingers width on each end	$3 * W_c$	3 μm	Min 600 nm	
Thickness of suspended parts	t	200 nm	200 nm	200 nm (part a)
Number of gaps in comb-drive	N	40	Min 2 Max 200	

A summary of the material and optical constants:

Young's modulus of Si	E_{Si}	170 GPa
Density of Si	ρ_{Si}	2330 kg/m ³
Light Wavelength	λ	1550 nm
Absorption constant	b	0.3 μm^{-1}
Effective refractive index at $g=0$	$\eta_{eff,0}$	2.6
Effective refractive index at $g \rightarrow \infty$	$\eta_{eff,\infty}$	2.4

Part a) 2D Model Optimization of optical phase shifter for quantum applications

1. The low-frequency modulation sensitivity (or $\left| \frac{\Delta\phi(s)}{V_{AC}(s)} \right|$ as $f_m \rightarrow 0$.) as a function of V_{DC} . The DC bias voltage in part 1 with maximum sensitivity was much higher than the derived pull-in voltage. This means, the device dimensions were not optimized for the practical use-case. Provide an analytical expression of V_{DC_MAX} .
2. Consider which terms of the expression found in a.1 are critical such that $V_{DC_MAX} < V_{Pull-In}$. Define these constraints and explain your reasoning.
3. Now, select new dimensions to optimize the maximum achievable modulation sensitivity with the geometry constraints provided above, and preserve following constraints of the system. You are free to use any optimization method or tool to find the best values. **Report your dimensions using the "Final Value" column on the table of page 2.**

Fundamental resonance frequency	f_r	>500kHz
Pull-in Voltage	$V_{Pull-In}$	> 200 V
DC Bias Voltage of maximum modulation sensitivity	V_{DC_MAX}	< $V_{Pull-In}$
Maximum AC acceleration that the system can withstand at the worst-case frequency.	$a_{0,max}(\omega_r)$	> $2 \times 10^4 \text{ m/s}^2$

4. Validate the new design in COMSOL and report the following parameters. Compare the values obtained from the COMSOL model with the analytical model predictions and discuss the deviations found and their possible causes
 - (i) Equivalent spring constant k ,
 - (ii) 2D eigenfrequencies maps of the 1st 5 modes
 - (iii) maximum DC displacement at the intended maximum voltage of operation,
 - (iv) maximum DC acceleration without hitting the waveguide
 - (v) The step response of the system and the rise time of the new device from a time domain analysis.

Part b) 3D model to mitigate out-of-plane behavior

Until now, we have been focused on 2D simulations of the optical modulator. Now, we will check in a 3D geometry simulation about the out-of-plane activity of the system. **For simplicity, we will use the device dimensions from part 2, and remove the comb drive fingers from the COMSOL geometry.**

1. The suspended part thickness t had been restricted to 200 nm in part 1, 2, and 3a. Explain why this could be problematic using your course knowledge from beam mechanics. (Hint: Calculate the equivalent spring constant for the out-of-plane direction compared to the in-plane.)
2. Now, find under what thickness t the desired resonance frequency relates to the fundamental eigenmode and natural frequency. Confirm your results with a 3D eigenfrequency analysis in COMSOL. Report the final dimensions, the equivalent spring constant, and the first 5 3D eigenmodes. Compare your results to analytical estimations and explain any discrepancies.